

Pathology Examination

Hemodynamic Disturbances

Thrombosis & Embolism

Part I: Multiple Choice Questions (MCQs)

1. According to Virchow's triad, a primary cause of thrombus formation is:

- A) Efficient collateral circulation
 - ✓ B) Endothelial cell injury
 - C) Patchy ischemic necrosis
 - D) Minimal reduction in blood supply
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2. In Virchow's triad, abnormal blood flow includes stasis and:

- A) Atherosclerosis
 - ✓ B) Turbulence (unsteady blood flow)
 - C) Neoplasia
 - D) Hypercoagulability
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3. A thrombus occurring in a non-inflamed vein (e.g. in heart failure or varicose veins) is called:

- A) Thrombophlebitis
 - ✓ B) Phlebothrombosis
 - C) Arteritis
 - D) Agonal thrombus
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4. The thrombus characterized by “Lines of Zahn” (alternating pale and dark layers) is:

- A) Pale thrombus
 - B) Red thrombus
 - ✓ C) Mixed thrombus
 - D) Capillary thrombus
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5. A pale (white) thrombus is mainly composed of:

- A) Fibrin with trapped RBCs
 - ✓ B) Platelets and fibrin (platelets > fibrin)
 - C) Bacteria and necrotic debris
 - D) Yellow fibrin only
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6. The usual fate of a septic thrombus is fragmentation leading to:

- A) Recanalization
 - ✓ B) Septic emboli and pyemic abscesses
 - C) Complete lysis
 - D) Calcification
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7. An embolus is best defined as:

- A) A mass formed during life
 - B) Impaction of a mass in a vessel
 - ✓ C) A detached intravascular mass circulating to a site distant from its origin
 - D) Granulation tissue invading a thrombus
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	<i>Thrombus</i>	<i>Clot</i>
Time of formation	Antemortem (during life)	Ant-mortem or postmortem
Location	intravascular	Extra and intravascular
Types of blood	Circulating	Stagnant
Attachment to vascular wall	Attached at certain site	Not attached
Composition	Platelets Fibrin RBC's and WBC's	<u>No platelets</u> Fibrin RBC's and WBC's
Lines of zahn	Present	Absent
Consistency	Friable	Elastic
Colour	Pale Red Mixed	Red yellow

Part III: Essay & Short Answer Questions

Q1. Fate of an Aseptic Thrombus

An aseptic thrombus (free of bacteria) may undergo the following fates:

- **Lysis:** Fibrinolytic activity dissolves the thrombus and restores blood flow
 - **Organization:** Granulation tissue invades the thrombus from the vessel wall
 - **Recanalization:** Formation of wide anastomosing channels through the thrombus
 - **Embolization:** Detachment of the thrombus forming an embolus
 - **Other outcomes:** Calcification, permanent vascular occlusion, or secondary infection
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Q2. Factors Causing Hypercoagulability (Virchow's Triad)

- **Genetic (Primary):**
 - Factor V Leiden mutation
 - Prothrombin gene mutation
- **Acquired (Secondary):**
 - Prolonged immobilization (bed rest)
 - Surgery or fractures
 - Pregnancy